STELLAR NUCLEOSYNTHESIS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 16, 2026

Diagram-based questions on Stellar Nucleosynthesis

Question 1 10.0 points

Analyze the proton-proton (p-p) chain reaction depicted in the diagram below for main sequence stars. Identify and explain the sequence of nuclear reactions occurring in the p-p chain, including the role of neutrinos and gamma rays in energy transport within the star.

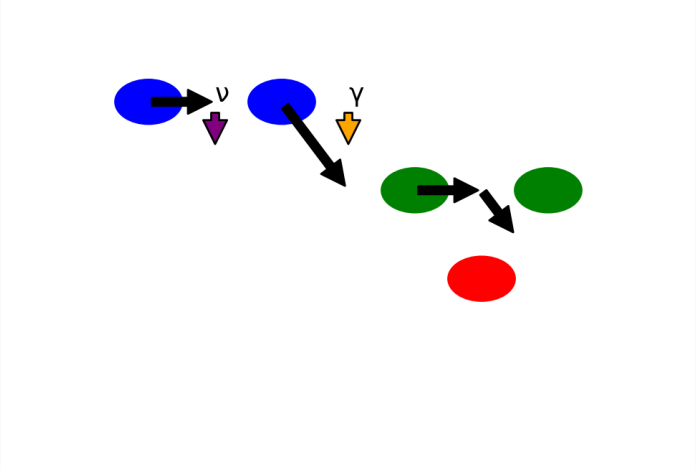

Fig. 1. Diagram of the proton-proton (p-p) chain reaction in main sequence stars. The diagram should show the sequence of nuclear reactions starting with two protons (hydrogen nuclei) fusing to form deuterium, followed by the production of helium-3, and finally the formation of helium-4. Include the emission of neutrinos and gamma rays at appropriate stages, with arrows indicating the direction of energy transport. Label each step of the reaction and the particles involved, such as protons, deuterium, helium-3, helium-4, neutrinos, and gamma rays. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • Two • Deuterium • Neutrinos The diagram will FAIL review if any of these appear.

Question 2 10.0 points

Examine the CNO cycle reaction network in the diagram below for massive stars. Describe how this cycle contributes to the stellar nucleosynthesis process and compare its efficiency to the proton-proton chain in terms of energy production.

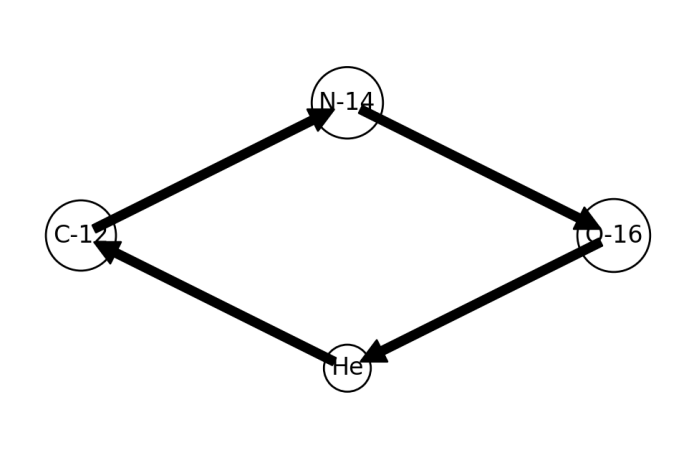

Fig. 2. A diagram of the CNO cycle reaction network for massive stars. This diagram should include the main isotopes involved: carbon-12 (C-12), nitrogen-14 (N-14), oxygen-16 (O-16), and their respective reactions. Arrows should indicate the flow of reactions, showing how hydrogen nuclei are converted into helium, with intermediate steps involving the capture of protons and the emission of positrons and neutrinos. Label each isotope and reaction step clearly, and include energy release at appropriate stages to illustrate the cycle's contribution to stellar nucleosynthesis. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • CNO The diagram will FAIL review if any of these appear.

Question 3 10.0 points

Interpret the Hertzsprung-Russell (H-R) diagram with evolutionary tracks shown in the diagram below. Explain how the diagram represents the life cycles of different mass stars and identify the key stages for a one-solar-mass star.

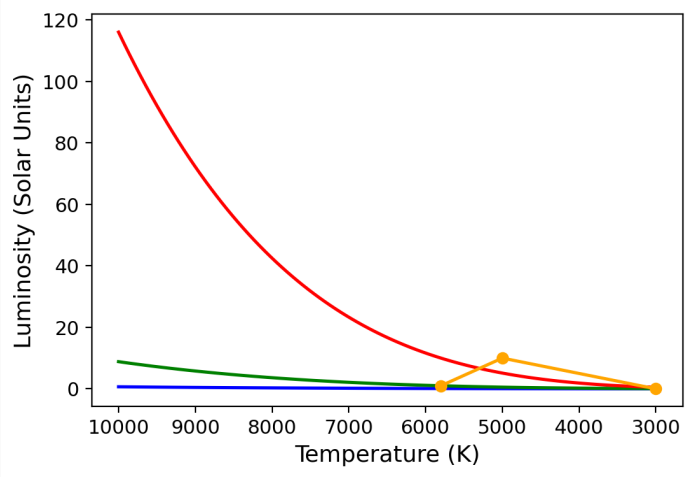

Fig. 3. Hertzsprung-Russell diagram showing evolutionary tracks for stars of different masses. The x-axis represents the temperature (decreasing to the right), and the y-axis represents luminosity. Include tracks for low, medium, and high-mass stars, highlighting the main sequence, red giant, and white dwarf stages. Specifically mark the evolutionary track of a one-solar-mass star and label key stages such as the main sequence, red giant branch, and white dwarf stage. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • key stages for a one-solar-mass star • logical qubit encoding path The diagram will FAIL review if any of these appear.

Question 4 10.0 points

Examine the binding energy per nucleon curve in the diagram below, focusing on the iron peak. Discuss how this curve illustrates the energy production limits in stellar fusion and why iron signifies an endpoint for nucleosynthesis in stars.

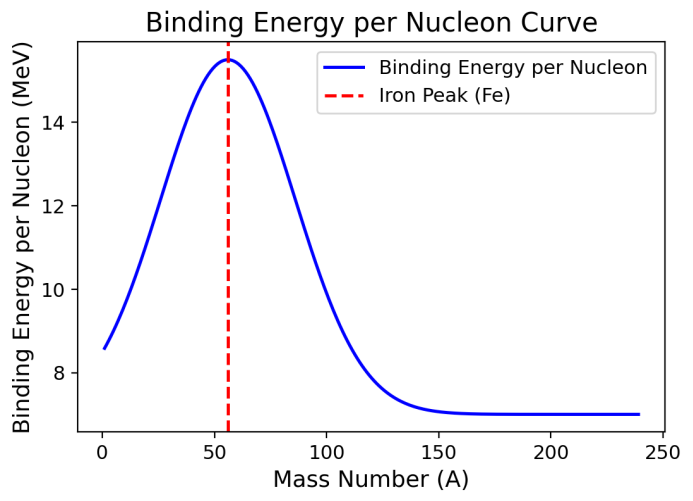

Fig. 4. Binding energy per nucleon curve showing the variation of binding energy with atomic mass number. The curve should highlight the iron peak, which represents the maximum binding energy per nucleon, and illustrate the general trend of increasing binding energy up to iron and then decreasing beyond it. This helps explain the energy production limits in stellar fusion and why iron is the endpoint for nucleosynthesis in stars.

Question 5 10.0 points

Examine the onion-shell model diagram of a pre-supernova massive star's internal structure shown below. Describe the different shells and their associated nuclear processes, explaining how this structure leads to a supernova event.

Fig. 5. Onion-shell model of a pre-supernova massive star. The diagram shows concentric layers representing different nuclear fusion processes. From the outermost to the innermost: 1) Hydrogen shell (H → He fusion), 2) Helium shell (He → C fusion), 3) Carbon shell (C → O fusion), 4) Oxygen shell (O → Ne fusion), 5) Neon shell (Ne → Mg fusion), 6) Magnesium shell (Mg → Si fusion), 7) Silicon shell (Si → Fe fusion), and 8) Iron core (no fusion, inert). Each shell is labeled with the element and fusion process occurring within it. The layers are arranged in decreasing order of size, with the iron core at the center. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • These • different shells The diagram will FAIL review if any of these appear.